MILESTONES IN THE EVOLUTION OF THE ATMOSPHERE WITH REFERENCE TO CLIMATE CHANGE

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ABSTRACT

The history of life on Earth is intertwined with, and critically dependent on, the carbon and oxygen cycles of the atmosphere-hydrosphere-lithosphere system. This included (1) periods of sustained high-greenhouse gas (GHG: CO_2, CO, CH_4, N-oxide) atmosphere; (2) periods of CO_2-sequestration by weathering, vegetation and burial of organic matter; (3) episodic injections of CO_2 from volcanic activity, from volatilized crust impacted by asteroids and comets, and from metamorphic de-volatilization processes, and (4) Quaternary cyclic high-amplitude orbital solar forcing (on a scale of 40 to 100 Watt/m^2 June insolation at latitude 65^{o}N)—triggering glacial albedo decline from melting ice sheets and CO_2-feedback effects from warming seas—resulting in glacial terminations. Lately a new factor emerged—a carbon-emitting warlike mammalian species. Since about 1750 ~305 Gigatons carbon (GtC) were combusted, compared to ~750 GtC in the atmosphere, about 10 percent of the known global fossil fuel reserve of ~4000 GtC, and about 12 percent of the terrestrial biosphere. The reserve compares with the release during the K-T impact event 65.10^6 years ago of ~ 4600 GtC to the atmosphere, associated with mass extinction of species. Since the 19th century, and in particular from the 1970s, the growth rate of atmospheric greenhouse gases and global mean temperatures exceeds those of Pleistocene glacial terminations by one to two orders of magnitude. The relations between temperatures and sea levels (SL) for the last few million years project future SL rises toward time averaged ratios in the order of 4 to 18 metre per 1 degree C. The observed ice melt lag effects of glacial terminations, spring ice melt dynamics in Greenland and west Antarctic, and the doubling per-decade of ice melt areas and sea level rises, suggest that the IPCC-2007-projected SL rises (~20-60 cm) for the 21st century may be greatly exceeded. The biological and philosophical raison d'etre underlying anthropogenically triggered climate change and mass extinction may never be known.

I. MILESTONES IN THE INTERTWINED EVOLUTION OF THE ATMOSPHERE-HYDROSPHERE-BIOSPHERE SYSTEM

The position which Earth occupies between Mars and Venus holds for the state of its atmosphere (mean T 18°C [-50 to +55]; 1 bar; N 79%; O_2 20%; CO_2 250–381 ppm; liquid water), intermediate between the thin atmosphere water-less state of Mars (-113°C–0°C; 0.01 bar; CO_2 95.3%; O_2 0.13%) and the greenhouse state of Venus (>450°C; 90 bar; CO_2 and SO_2). Secular variations in atmospheric levels of carbon dioxide, carbon monoxide, methane, oxygen, nitrogen oxides, Sulphur dioxide and dust with time manifest the intimate interdependence of climate states on the physical and chemical parameters of the atmosphere and oceans, on continent-ocean geotectonic patterns, mountain belts, ice sheets, vegetation and life forms (Frakes et al., 1992; Beerling et al., 2002; Berner, 2004, 2006; Beerling and Berner, 2005, 2006; Royer, 2006; Berner and Beerling, 2007; Berner et al., 2007; Royer et al., 2007).

The domination on the early Earth of high geothermal gradients and of greenhouse conditions, induced by release of carbon and Sulphur volatiles from volcanic activity and asteroid/comet impact-volatized crust, and of low oxygen levels evidenced by unoxidized pyrite and uraninite in sediments, ensued in a methane-rich and possibly CO-rich atmosphere and in warm low-pH oceans of low CO_2 solubility. Low oxygen conditions are further evidenced by the abundance in 3.85 – 1.8 Ga terrains of banded iron formations, reflecting abundance of soluble ferrous iron in sea water (Trendall and Blockley, 1970; Cloud, 1973; Morris, 1993), Whereas at present atmospheric CO levels vary between 0.01 – 2.0%, under an unoxic Archaean atmosphere higher CO and CH4 levels resulted in dominance of methanogenic microbes in acid anoxic “purple” oceans inhabited by iron and sulphur metabolizing microbes (phototrophic purple Sulphur bacteria [Chromatiaceae] and green Sulphur bacteria [Chlorobiaceae]) (Brocks et al., 2005).

Eclaves of cherts in the 3.5–3.2 Ga Swaziland Supergroup, Barberton greenstone belt, South Africa, display low $\delta^{18}O$ values indicating extremely high ocean temperatures of 55–85°C (Knauth and Lowe, 2003), limiting the solubility of CO_2. The relatively small (compared with present continents) size of Archaean continental nuclei relative to oceanic domains (McCulloch and Bennett, 1995; Glikson, 2001) reduced rock exposure, weathering and CO_2 sequestration. The lack of land vegetation and thus lesser degree of leaching of Ca and Mg to precipitate carbonate, and the absence of carbonate-accreting marine plankton, reduced CO_2 sequestration, as reflected by a relative scarcity of carbonate sediments.

The origin of Precambrian ice ages, including the Huronian (~2.4-2.1 Ga) and the Cryogenian (850-630 Ma Sturtian and Marinoan “snowball Earth” glaciations, reaching a peak at 607.8±4.7 Ma) (Kirschvink, 1992; Hoffman et al., 1998; Keller, 2005), is unknown. Possibilities include as episodic oxidation of methane-rich greenhouse atmosphere, reducing its greenhouse intensity, or increased weathering sequestration by metamorphism and uplift of Pan African mobile belts. The appearance of multicellular Ediacaran fauna in the wake of the glaciation may be related to lowered CO_2 and elevated oxygen levels (Canfield et al., 2007).

Phanerozoic CO_2 and O_2 levels and palaeo-temperatures reveal a dominance of greenhouse conditions in the early Palaeozoic (~500 Ma, >4000 ppm CO_2, +7°C APL [mean temperature above present levels]). The appearance of the first land plants in the upper Silurian (~420 Ma) was associated with a marked decline in CO_2 levels and temperature, and an increase in oxygen (from ~15 to ~25% O_2) reaching glacial conditions during the late Carboniferous-early

Permian (Figs 1-3). The advent of land plants resulted in a coupled evolution of vegetation and the carbon and oxygen cycles heralded a fundamental change in the nature of the atmosphere-hydrosphere system (Beerling and Berner, 2005). Positive CO_2 feedbacks arise by warming seas and release of CO_2 and methane from drying bogs and sediments and from burning vegetation. Negative feedbacks include absorption of CO_2 by plants and soil, burial of fossil organic matter, chemical weathering and leaching of soluble Ca and Mg from rock and soil enhanced by plants, with subsequent precipitation of carbonate. Increased O_2 and decreased CO_2 in the atmosphere results in gigantism in plants aimed at absorbing sufficient CO_2, and in animals as an effect of excess O_2 (Berner et al., 2007).

The removal and burial of atmospheric CO_2 by plants during the Carboniferous due to burial of organic matter, especially in the form of microbially resistant lignin, led to a large drop in atmospheric CO_2, reducing greenhouse effect and setting the stage for the vast Permo-Carboniferous glaciation (Fig. 1). Increased burial of carbon-rich matter resulted in the increased production of O_2, reaching levels as high as 30% in the Permo-Carboniferous (Fig. 2). This elevated O_2 likely contributed to an increase in the size of organisms, such as insects and amphibians, which breathe via diffusive processes. The effects of massive volcanism, CO_2 release, low-pH oceans and anoxia, associated with the Permian-Triassic boundary at ~251 Ma, saw a demise in vegetation and of carbon burial, thereby a decrease in O_2 production (Fig. 3) (Keller, 2005).

Greenhouse Earth conditions persisted through the upper Mesozoic, with occasional glacial breaks such as in the Jurassic. Given marked temperature variations between the Jurassic (minimum mean temperatures similar to the present) and the mid to upper Cretaceous (maximum mean temperatures about 4^oC higher than at present), the ~200 million years-long survival of the dinosaurs is in part attributable to the development of specialized bird-type breathing apparatus.

II. ATMOSPHERIC EFFECTS OF VOLCANIC AND ASTEROID IMPACTS

Several Phanerozoic mass extinctions temporally associated with volcanic and extraterrestrial impact events (Glikson, 2005; Keller, 2005) were likely closely related atmospheric changes in CO_2 and oxygen levels, with long term effects due to feedback effects from oceans and the biosphere, including the following:

1. Mass extinction (~42 percent of genera) in the mid-Cambrian ~513 Ma associated with the eruption of the Antrim Plateau Basalts (507-513 Ma).
2. Mass extinction (~58 percent of genera) in the late Devonian (Frasnian-Fammenian boundary ~374 Ma and end-Devonian ~359 Ma) is associated with a cluster of large asteroids (Woodleigh, D_{crater} ~120 km; Alamo, ~100 km; Siljan, ~52 km; Charlevoix, ~54 km, and a number of smaller impacts) (Glikson, 2005), accompanied with high CO_2 levels (~2000-3000 ppm), high temperature (+2-5^oC APL) and low Oxygen (~14-15%) (Figs 1-3).
3. The eruption of extensive Siberian basalts at ~251 Ma (Changhsingian) associated with catastrophic changes in the atmosphere and hydrosphere, triggering the largest extinction in Earth history (~78 percent of genera, Keller, 2005). According to R. Berner (pers. comm., 2007) a CO_2 double spike of >20,000 ppm occurred due to volcanic gas and sharp reduction in vegetation and burial rates of organic matter (Fig. 4). Mean

values across the boundary include CO_2 levels of up to 2000 ppm, high temperature (+5°C) and low oxygen levels (15-20%) (Fig. 2).

4. Sharp CO_2 spikes (up to 4000 ppm) and low oxygen (~15% O_2) associated with the end-Triassic (~200 Ma) (Norian) are attributed to extensive volcanic activity associated with opening of the incipient Atlantic Ocean ~200-202 Ma (Berner and Beerling, 2007). These authors estimate release of 21,000 GtC as CO_2 and 57,000 GtS as SO_2, resulting in low pH and consequent undersaturation of the oceans in calcium carbonate (Fig. 5). Mass extinction of 35 percent of genera occurred ~ 216-203 Ma and could have been enhanced by large extraterrestrial impacts (214 Ma; Manicouagan, 100 km, Rochechourat, 23 km)
5. The end-Jurassic ~145 Ma was associated with a major asteroid cluster impact cluster (Morokweng, >90 km; Mjolnir, 40 km; Gosses Bluff, 24 km). Mass extinction reached 20 percent of genera (Tithonian).
6. The Cretaceous-Tertiary boundary Chicxulub-Boltish impact cluster (KT impact) (Alvarez et al., 1980; Keller, 2005). According to Beerling et al. (2002) late Cretaceous/Early Tertiary background CO_2 levels of 350-500 ppm peaked at 2300 ppm within 10,000 years of the KT impact, due to instantaneous transfer of approximately 4600 GtC to the atmosphere by the KT boundary impact. Preceding and accompanying Deccan Plateau volcanism contributed to a CO_2 and SO_2–rich atmosphere. A resultant climatic forcing of +12 W/m2 would have been sufficient to warm the Earth's Surface by approximately +7.5°C, in the absence of counter forcing by sulfate aerosols. However, the presence of evaporite sediments in the Chicxulub crater target rocks suggests release of large volumes of sulfate as well as dust, resulting in a short "impact winter" (Toon et al., 1997). Extinction of genera associated with these events reached 48 percent (Keller, 2005).

Based on current climate sensitivity measurements (Hansen et al., 2007) atmospheric CO_2 rises result in temperature increase on the scale of ~1°C per 100 ppm CO_2, implying many degrees rise for GHG release in the PT and KT events. No correlations between mass extinctions at the end-Cambrian (~488 Ma; ~40 percent of genera) and the end-Ordovician (444 Ma; ~60 percent of genera) and major volcanic or extraterrestrial impacts are evident from Keller's (2005) compilation.

III. FROM A TERTIARY GREENHOUSE WORLD TO QUATERNARY ICE AGES

Zachos et al. (2001) summarized deep-sea sediment core evidence for Cainozoic climate changes, involving (A) gradual trends of warming and cooling driven by tectonic processes on time scales of 10^5 to 10^7 years; (B) rhythmic or periodic cycles driven by orbital processes on time scale of 10^4- to 10^6-year, and (C) rare rapid climate shifts with durations of 10^3 to 10^5 years. A sharp rise in the mean global temperature following the K-T boundary impact, possibly reflecting long term effects of this event, was punctuated at the Palaeocene-Eocene boundary (55 Ma) by a release of methane hydrates from ocean sediments (Fig. 6). Mass balance calculations indicating a decline in d^{13}C in the order of 2.5‰ suggest addition of ~1500 GtC of isotopically light carbon (d^{13}C -60‰) to the atmosphere and oceans, raising CO_2 levels to 900 ppm (Gingerich, 2006). As methane has X23 the greenhouse effect of CO_2, its release resulted in an increase in d^{18}O of 1-2‰, suggesting the deep and high latitude oceans warmed by ~7 - 8°C and mid-latitude terrestrial temperatures by ~4 - 5°C (Zachos et al, 2001; Wing et al., 2005; Gingerich, 2006; Higgins et al., 2006). This was accompanied with a mass extinction of benthic forams and, at the same time, diversification of continental mammals.

A Quaternary greenhouse event is documented in the mid-Pliocene (~3 Ma), when temperatures rose about 2-3°C higher than at present and sea levels by about 25±12 metres (Fig. 7) (Dowsett et al., 1994), as documented for example in Florida. Estimates of sea surface temperature based upon foraminifera, diatom, and ostracod assemblages from ocean cores reveal a warm phase of the Pliocene between about 3.3 and 3.0 Ma (Dowsett et al., 2005). Pollen records and plant megafossils, although not as well-dated, show evidence for a warmer climate at about the same time. Increased greenhouse forcing and altered ocean heat transport are the leading candidates for the underlying cause of Pliocene global warmth. The data suggest long-term stability of low latitude sea surface temperatures and greater variability in regions of maximum warming.

Quaternary glacial-interglacial cycles triggered by orbital forcing (Milankovic, 1941; Imbrie and Imbrie, 1980), including eccentricity (~100 and ~400 kyr periods), obliquity (41 kyr), and climatic precession (~19 and ~23 kyr), consistent with d^{18}O of marine sediments and ice cores (Hayes et al., 1976), result in insolation variations and ice albedo changes in June insolation at latitude 65°N of 40 to 100 Watt/m^2 (Roe, 2006) (mean global insolation changes are significantly lower). Greenhouse gas feedback effects result in secondary variations on the scale of ~2 Watt/m^2. (Roe, 2006), which explains the lag of CO_2 rises behind temperature rises by an average of 700 years (Hansen et al., 2007). A close correlation ensues between insolation rises, decrease in ice volumes (Fig. 8) and increase in the time rate of ice sheet volume changes (dv/dt), where sea level rise lags by a mean of 1 kyr behind ice sheet melt (Roe, 2006).

Palaeo-sea level studies (Siddall et al., 2003, 2006) (Fig. 9) and temperature and GHG glacial-interglacial age profiles (Petit et al., 1999; EPICA, 2004; Siegenthaler et al., 2005; Hansen et al., 2007) document sea level rise of +2 to +4 metre higher than at present at Termination II (130-122 kyr) (Fig. 10). Others suggest sharp sea level rise to +6 to +8 metres and a short high-stand for ~600 years on the basis of wave cut notches, rubble benches and shoreline sediments (Hearty and Neumann 2001), consistent with partial collapse of the Greenland and West Antarctic ice sheets. This would have involved a flux of fresh water and disruption of the thermohaline circulation (Hansen et al., 2007). Following peak temperatures at ~130–126 kyr, sea levels continued to rise between 124.5-122.5 kyr (Fig. 10; Table 2). The overall mean global temperature rise associated with Termination-II from ~130 kyr, based on the Vostok ice core (Petit et al., 1999), is about +4.5°C (half the Vostok anomaly).

Comparison between Siddall et al.'s (2005) sea level data and Petit et al.'s (1999) Vostok ice core palaeo-temperature data indicate sea level rise lag effects following stabilization of temperatures at Termination-I (~11.5 kyr, "Younger Dryas") continued for a further 4000 years, raising sea levels by 40 meters by 7 kyr (Fig. 11). Tentative evidence suggests a large asteroid impact coincides with this period (Dalton, 2007)

IV. POLAR ICE MELTING AND SEA LEVEL RISE

The IPCC-2007 report indicates an overall increase in the rate of sea level rise from 0.18±0.05 cm/year in 1961--2003 to 0.31±.07 cm/year in 1993-2003, including contributions from combined doubling of Greenland and Antarctic ice sheets melt from 0.019 cm/year to 0.041 cm/year between these periods (Table 1A), related to mean global temperature increases of 0.3°C and 0.6°C, respectively. According to the IPCC-2007 "The remainder of the ice loss from Greenland has occurred because losses due to melting have exceeded accumulation due to snowfall". Due to sparse distribution of measurement stations in Antarctica the balance between ice melt and snow fall is not clear.

NASA satellite gravity and microwave measurements indicate a doubling of Greenland ice melt areas per-decade (NASA 2006). Rates of ice loss of the Greenland ice sheet have increased from 0.05±0.12 mm/yr during 1961--2003 to 0.21±0.07 mm/yr during 1993--2004. The measurements indicate an increase in ice sheet melt area by 16% from 1979 to 2002 (Steffen and Huff, 2002; Steffen et al., 2004; NASA, 2006; Hanna et al., 2005; Thomas et al., 2006; IPCC-2007; Hansen et al., 2007) (Fig. 12).

Time-variable gravity measurements from the GRACE (Gravity Recovery and Climate Experiment) satellites of mass variations of the Antarctic ice sheet during April 2002–August 2005 found that the mass of the ice sheet decreased at a rate of 152±80 cubic kilometres of ice per year, equivalent to 0.4±0.2 mm of global sea-level rise per year. Most of this mass loss came from the west Antarctic Ice Sheet, including a water equivalent decrease in ice thickness of -1 to -4 cm/year for the Antarctic peninsula and the Ross Sea-Amundsen Sea shelf area (Rignot and Thomas, 2002; Chen et al., 2006; Velicogna and Wahr, 2006) (Fig. 13). GRACE-based estimates by Chen et al. (2006) identify ice loss of 77±14 km^3/year in West Antarctica and gain of +80±16 km^3/year in Enderby Land of East Antarctica.

The IPCC-2007 WG-1 report estimates a sea level rise of 18–59 cm by 2100, including caveats regarding millennia-scale melting of the Greenland ice sheet and the unknown nature of ice melting dynamics. Rahmstorf (2006) used linear extrapolation of relations between decadal sea level rise and global warming, suggesting a proportionally constant value of 0.34 cm/year per +1^{o}C temperature rise, projecting sea level rises in 2100 of +0.5--1.4 metres above 1990 level. Nonlinear doubling of combined Greenland and Antarctic 1993-2003 melt rates of +0.41 mm/year could lead to exponential sea level rise reaching levels reaching ~0.5 – 0.6 metre by mid-century and ~4.6 metres toward the end of the 21st century (Table1B; Fig. 14)—consistent with warnings by Hansen et al. (2007A,B).

The above estimates do not take into account potential factors such a sharp decrease in ice sheet albedo, collapse of the Gulf Stream and changes in the thermohaline circulation (Hansen et al., 2007; Pittock, 2007). Advanced melting of Arctic sea ice documented between 1950--2005 (US National Centre Arctic Research) reduced the extent of September sea ice from ~8.5 to ~5.0 million km^2, with significant effects on the Earth's albedo. Amplification of anthropogenic GHG feedback effects by release of methane hydrates from bottom sediments of warming seas, thawing permafrost and tropical bogs, and CO_2 release from drying vegetation and soil, extensive bush fires in tropical regions, threaten higher order atmospheric and sea level effects, approaching yet undefined tipping points (Hansen et al., 2007a,b; Glikson, 2007; Pittock, 2007)

Observed temperature rises through the 21st century were decoupled from solar insolation temperature effects about the mid-1970 (Fig. 15), and are tracking at the top of ICPP-2007 predictions (Fig. 16). A time-averaged plot of sea level vs temperature suggests a range from minimum values of ~4 metre/1^{o}C to maximum value of ~18 metre/1^{o}C (Rahmstorf et al., 2007 [Fig. 17]). If current GHG rise rates and Ice sheet melt rates are an indication, the millennia-long delays in sea level rise shown by Termination-I (Fig. 11) may be exceeded.

V. TERMINATION ZERO?

Based on palaeo-sea level and ice core palaeo-temperature studies (Siddall et al., 2003, 2006; Petit et al., 1999; Siegenthaler et al., 2005; EPICA, 2004) the relations between GHG forcing, temperature and sea level rises during Termination-II (~130-122 kyr), Termination-I (11.5 - 8 kyr), and since the mid-19th century, referred to here as "Termination 0", vary by large factors to orders of magnitude (Table 2):

1. Rates of CO_2 rise since the 1970s (1970-2003 1.57 ppm/yr) are higher than during 1850-1970 (0.42 ppm/yr) and two orders of magnitude higher than at Termination-I (0.014 ppm/yr) and termination-II (0.01 ppm/yr) (Table 2). CO_2 levels rose by over 30 percent from the maximum levels of ~290 ppm during the last 740 kyr to ~380 ppm in 2006.
2. The ratio of temperature rise per 1 ppm CO_2 during 1970-2003 (0.011^{o}C/1 ppm CO_2) has increased since 1850-1970 (0.0016 ^{o}C/1 ppm CO_2) by a large factor. These values are lower than Termination-I (0.15 ^{o}C/1 ppm CO_2), suggesting possible lag effects.
3. Rising sea level rates (from 0.11 cm/yr during 1870-1970 to 0.42 during 1970-2003) are between a fifth and a quarter of Termination-I. Metre-scale sea level rises can be expected if temperatures track the high-range estimates of 1.8-4.0^{o}C toward end-21st century (Houghton, 2001; IPCC-2007), as observed (Rahmstorf et al., 2007) (Fig. 16).

The long residence time of carbon dioxide in the atmosphere (5–200 years) and the lag effects of ice cap melting are matched by the inertia of the carbon-emitting species Homo sapiens. With CO_2 levels rising above 400 ppm at rates higher than 2 ppm/year, it is likely global warming of at least another +1^{0}C is already on track (Hansen, 2007A, B; Pittock, 2007; Rahmstorf. 2007), compounded by yet little defined feedback effects from warming oceans, thawing permafrost, and major albedo reductions associated with melting ice sheets and reduced aerosol levels -- resulting in a new atmospheric state.

Ever since discovering fire some 700 kyr ago, Homo *erectus* has been gaining increasing effect over its environment - justifying its re-definition as Homo *Prometheus*. The biological and philosophical question of the natural raison d'etre of anthropogenically triggered climate change and mass extinction defy understanding. Metaphorically, having reached the pinnacle of intelligence and consciousness, the species is paying the price for forbidden insights into subatomic matter, the Universe, time and space—perpetrating the next mass extinction.

Human inertia is ill-tuned to major shifts in atmospheric parameters. Present in societies are vested interests, ideologies promoting open ended material growth, survivalist paradigms, dreams of space colonization indifferent toward nature and fundamentalist cravings for the *rapture*. Arguments raised by "climate skeptics" (so-called) are reminiscent of the "God in the gaps" quest creationists direct in their criticism of Darwinian evolution. It is likely last resort geo-engineering efforts, aimed at gaining time, may be undertaken (Crutzen, 2006), with rather unpredictable consequences (Monbiot, 2006). Humans possess the ingenuity and technology needed to arrest dangerous climate change—provided they act in time.

Acknowledgements

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Source of sea level rise	Rate of sea level rise (mm per year) 1961 – 2003	1993 – 2003
Thermal expansion	0.42 ± 0.12	1.6 ± 0.5
Glaciers and ice caps	0.50 ± 0.18	0.77 ± 0.22
Greenland ice sheet	0.05 ± 0.12	0.21 ± 0.07
Antarctic ice sheet	0.14 ± 0.41	0.21 ± 0.35
Sum of individual climate contributions to sea level rise	1.1 ± 0.5	2.8 ± 0.7
Observed total sea level rise	1.8 ± 0.5[a]	3.1 ± 0.7[a]
Difference (Observed minus sum of estimated climate contributions)	0.7 ± 0.7	0.3 ± 1.0

Table 1A – Observed rate of sea level rise and estimated contributions from different sources (IPCC-2007 Table 5.3)

Table 1B. Projected decadal sea level rises through the 21st century[1] based on assumed doubling per-decade of ice sheet melt areas and mountain glaciers (Steffen and Huff, 2002; Steffen et al., 2004; NASA, 2006; Hanna et al., 2005; Thomas et al., 2006; IPCC-2007; Hansen et al., 2007).

mm	1993-2003 mm/ 10 yr[1]	2003-2013 mm/ 10yr	2013-2023 mm/ 10yr	2023-2033 mm/ 10yr	2033-2043 mm/ 10yr	2043-2053 mm/ 10yr	2053-2063 mm/ 10yr	2063-2073 mm/ 10yr	2073-2083 mm/ 10yr	2083-2093 mm/ 10yr	Cumulative 2003-2093
Greenland ice melt [2]	2.1	+4.2	+8.4	+17	+33	+67	+134	+267	+534	+1069	**2134**
Antarctica Ice melt [2]	2.1	+4.2	+8.4	+17	+33	+67	+134	+267	+534	+1069	**2134**
Glaciers & ice cap [3]	7.7	+15	+31	+62	+123						**225**
Thermal Expansion [2]	16 mm (per 0.33^{o}C)	+16	+16	+16	+16	+16	+16	+16	+16	+16	**144**
Total	27.9 mm	39.4	63.8	112	205	150	284	550	1084	2154	**4.64 metre**

[1] Mean values in mm/year based on IPCC-2007, Table SPM-1; [2] Assuming doubling of melting of Greenland and west Antarctic ice sheets and of mountain glaciers, and exhaustion of the latter by mid-century (no values are added from 2043); [3] Thermal expansion during the 21st century assume linear temperature rise.

Table 2

Period	mean CO_2 ppm	CO_2 rate ppm/yr	Mean CH4 ppb	Mean T°C	T°C / 1 ppm CO_2	Sea Level (cm)	SL cm/yr
A. Anthropocene/Holocene							
1970-2003	~325-377 [2]	2.2	1400-1750 [2]	13.9-14.5 [1]	0.011	**1970-2003** + 14 cm [3]	0.42
1850-1970	280-330 [1]	0.42	750-1400 [2]	13.7-13.9 [1]	0.0016	**1870-1970** + 11 cm [3]	0.11
10kyr-1750	265-285 [1]	0.002	~700 [2]			10-9 kyr ~2000 cm [3] 9kyr-1750 2200 cm [3]	2.0 0.25
B. Glacial Termination-I (values from Petit et al., 1999; Siddall et al., 2003)							
11.5-8.5	~265-260	Decline In CO_2	~600-570	+1.0		-62 to -12 (+50)	1.6 cm/yr
14–11.5	~235-265	**17-11.5kyr** 0.014 ppm/yr	~450-600	+4.5	0.15	-95 to -62 (+33)	1.32 cm/yr
C. Glacial Termination-II (Petit et al., 1999; Siddall et al., 2003)							
130-128	~190-290	**139-130kyr** 0.011 ppm/yr **130-120kyr** decline	~320-720	+6°C	0.06		
124.5-122.5	Intermit. decline		Intermit decline	decline		+32m (inc.+12m above present sea level	1.6 cm/yr

[1] mean global T (IPCC_-2007); [2] Mouna Loa (USGS); [3] Siddall et al., 2003

Main references: Petit et al., 1999 ; Siddall et al., 2006 ; IPCC-2007; Holgate, 2003

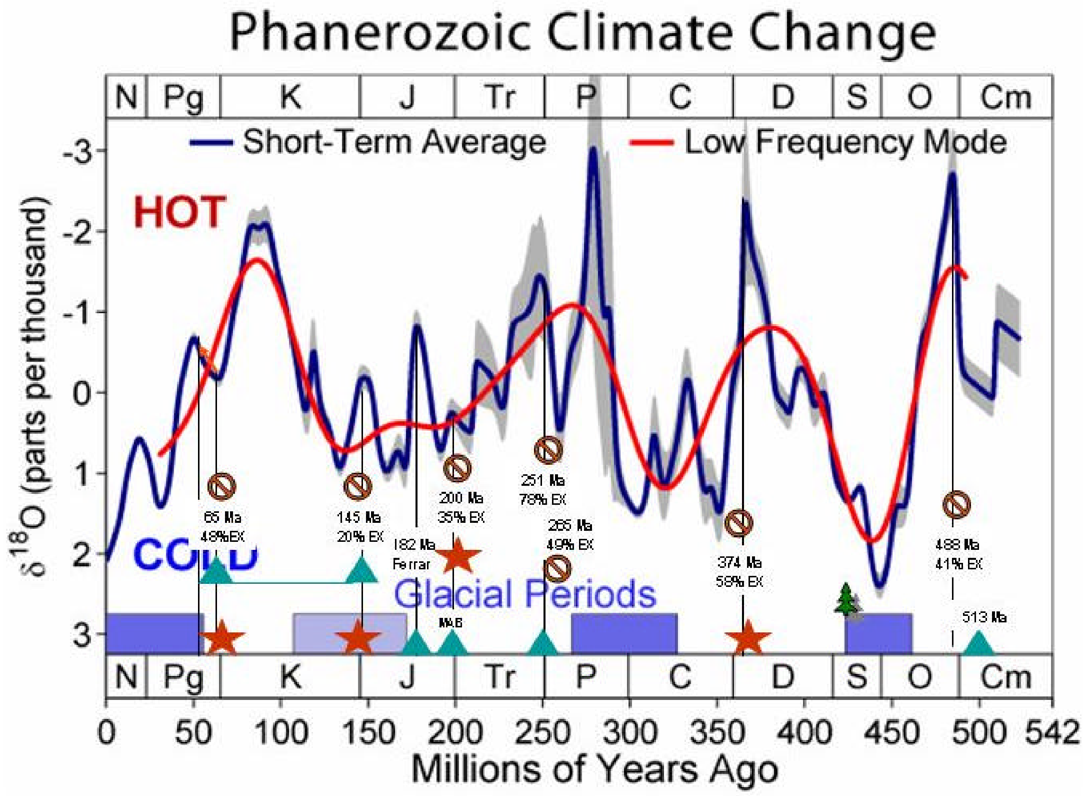

Figure 1. Phanerozoic climate change diagram (Wilkepedia) showing short term and low frequency mean temperature proxy variations ($d^{18}O$), glacial periods, major volcanic events (triangles), asteroid impacts (stars) and mass extinctions (bar signs, specifying age and percent extinction of genera)
(http://en.wikipedia.org/wiki/Geologic_temperature_record)

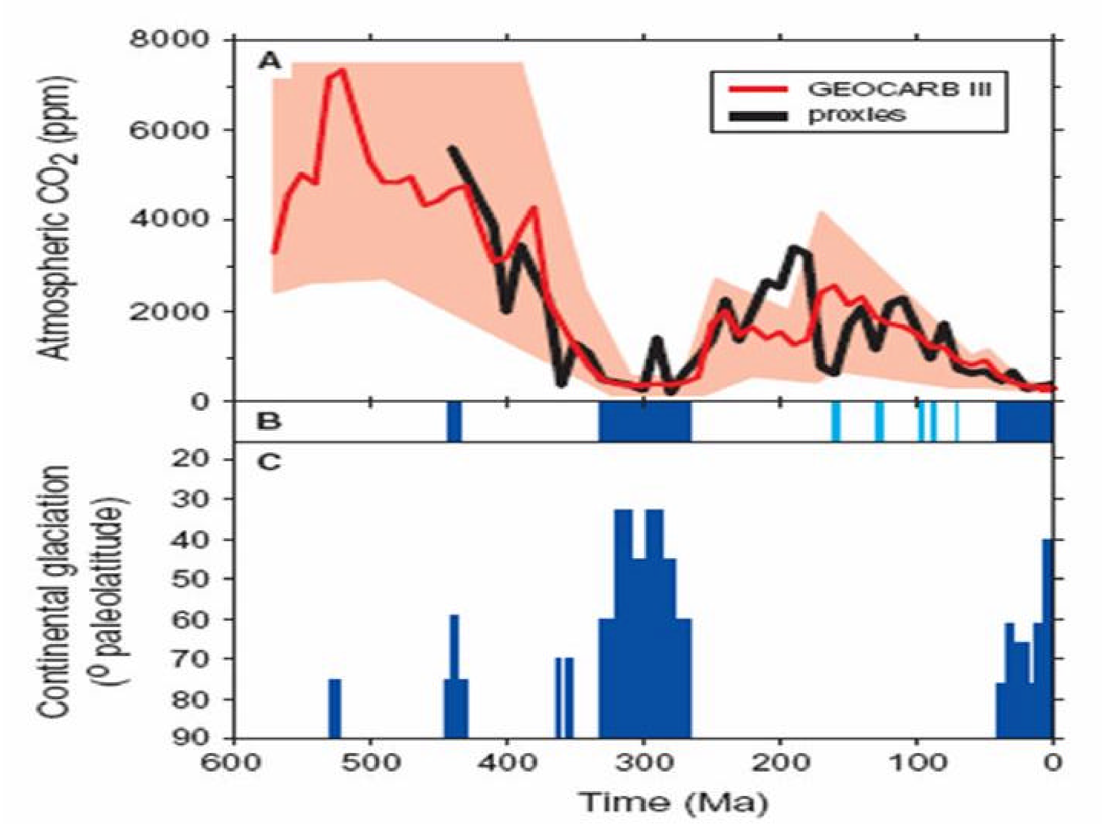

Figure 2. CO2 and climate. **A:** Comparison of model predictions (GEOCARB III) and proxy reconstructions of CO2. 10 m.y. time-steps are used in both curves. Shaded area represents range of error for model predictions. **B:** Intervals of glacial (dark blue) or cool climates (light blue; see text). **C:** Latitudinal distribution of direct glacial evidence (tillites, striated bedrock, etc.) throughout the Phanerozoic (Royer et al., 2004).

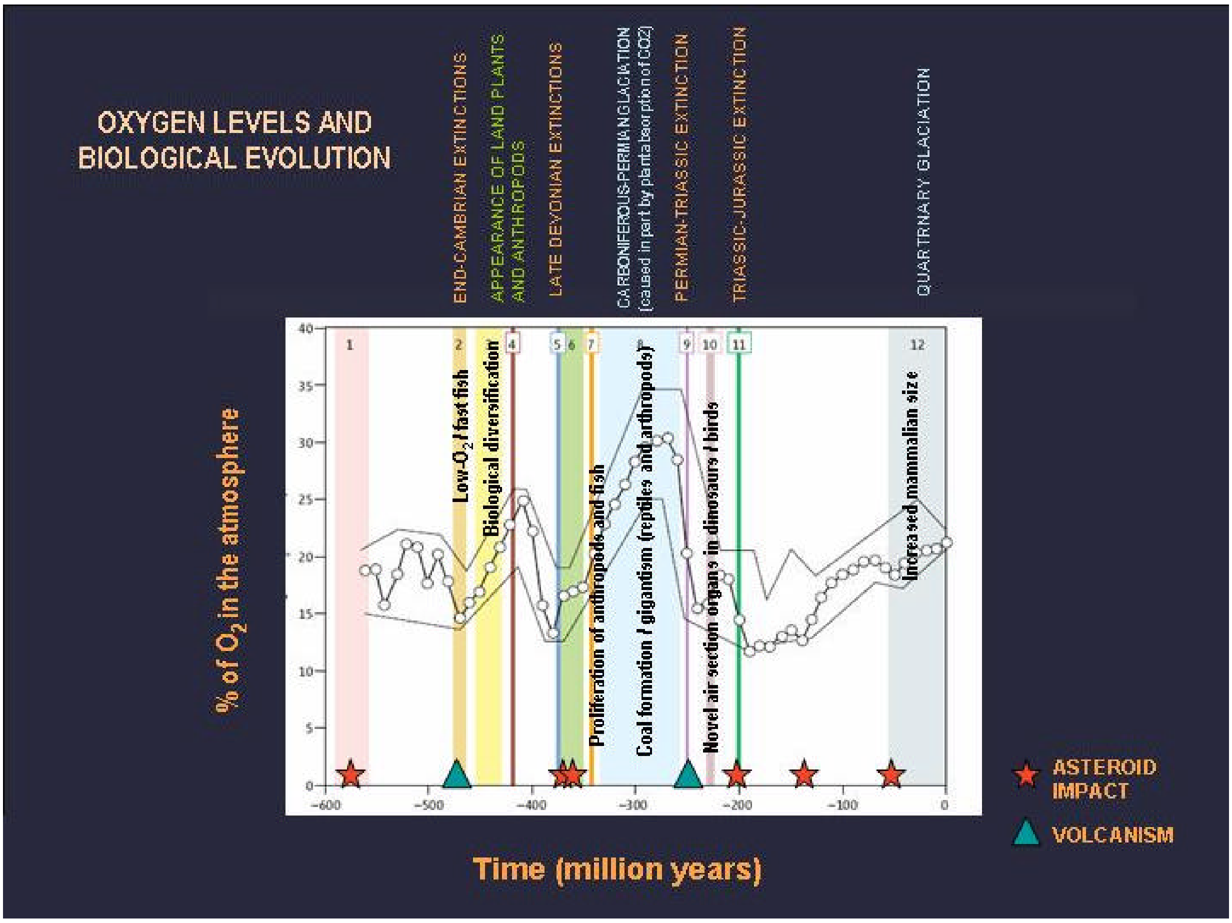

Figure 3. Atmospheric O_2 variations through the Phanerozoic (after Berener, 2006). The upper and lower boundaries are estimates of errors in modeling atmospheric O_2 concentrations. The numbered intervals denote evolutionary events that may be linked to changes in O_2 concentrations. Annotations by the author.

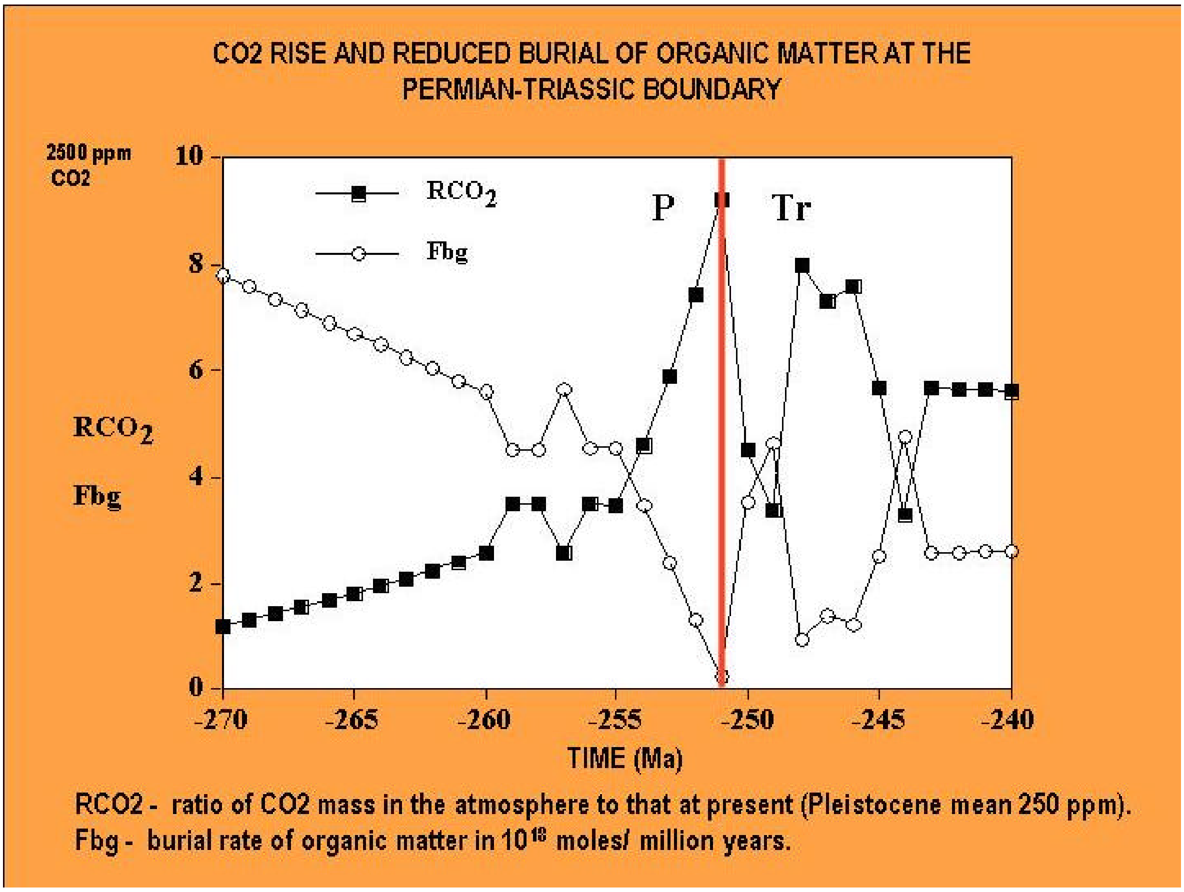

Figure 4. Variations in R_{CO2} (CO2 levels relative to the Pleistocene mean of 250 ppm) and of the burial rate of organic matter across the Permian-Triassic boundary (251 Ma) showing a spike in CO2 (near 2500 ppm) and sharp reduction in organic activity (R. Berner, pers.com., 2007).

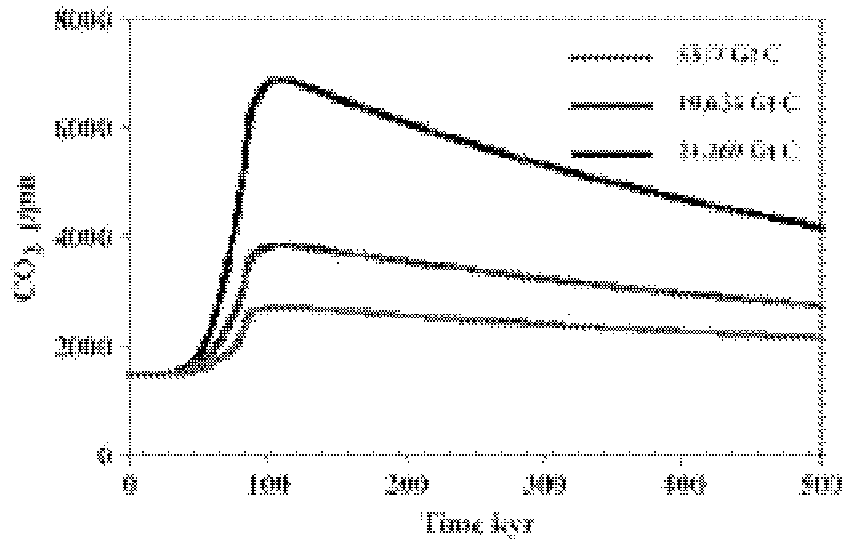

Figure 5. Effect of total degassed volcanic CO2 (in GtC) on atmospheric CO2 over 100 years with SO2/CO2 = 1; B. Effect of CH4 degassing on atmospheric CO2 and on carbonate saturation (Berner and Beerling, 2007).

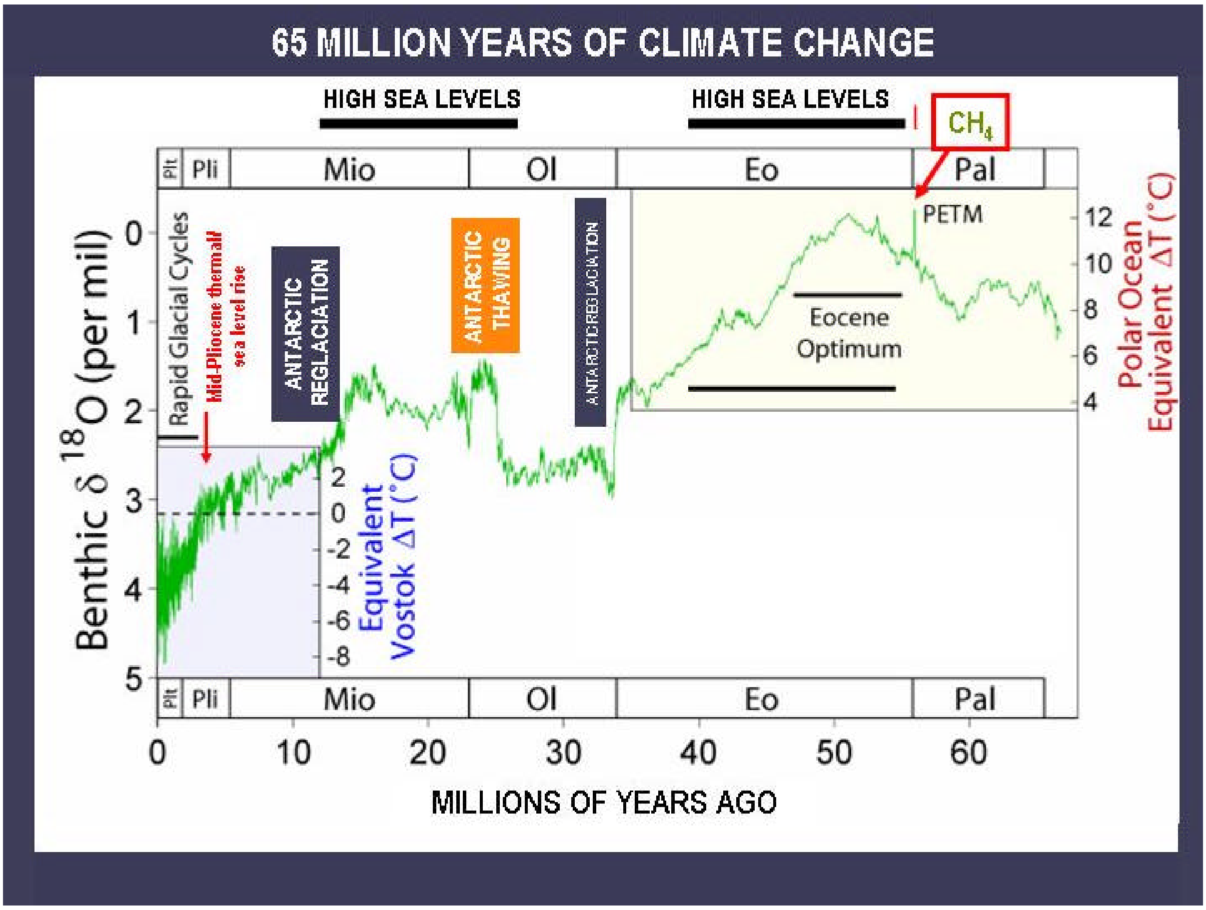

Figure 6. 65 million years of climate change based on sedimentary and isotopic studies of marine sediments (Zachos et al., 2001). Annotated by the author.

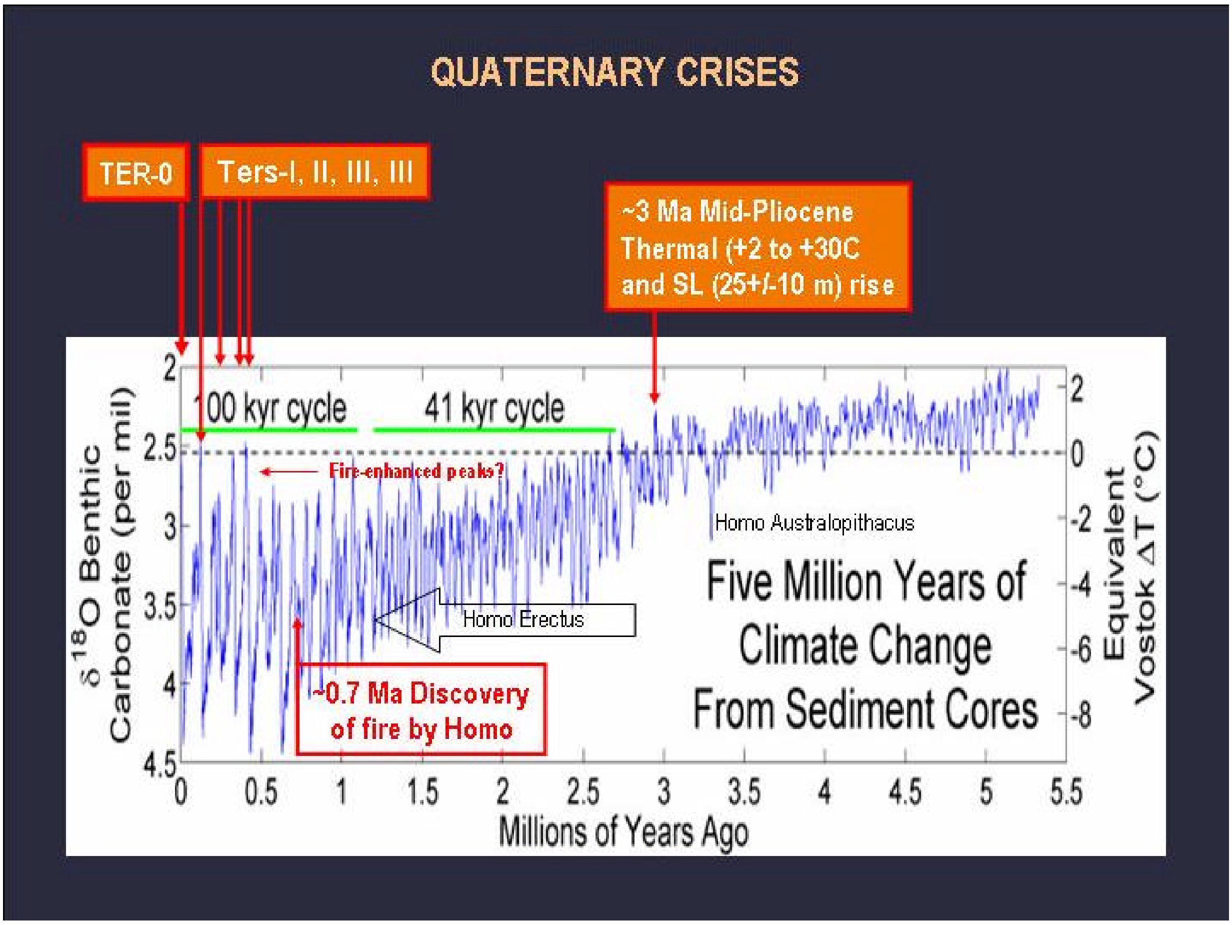

Figure 7. Five million years of climate change. Annotated by the author. (http://en.wikipedia.org/wiki/ Geologic_temperature_record)

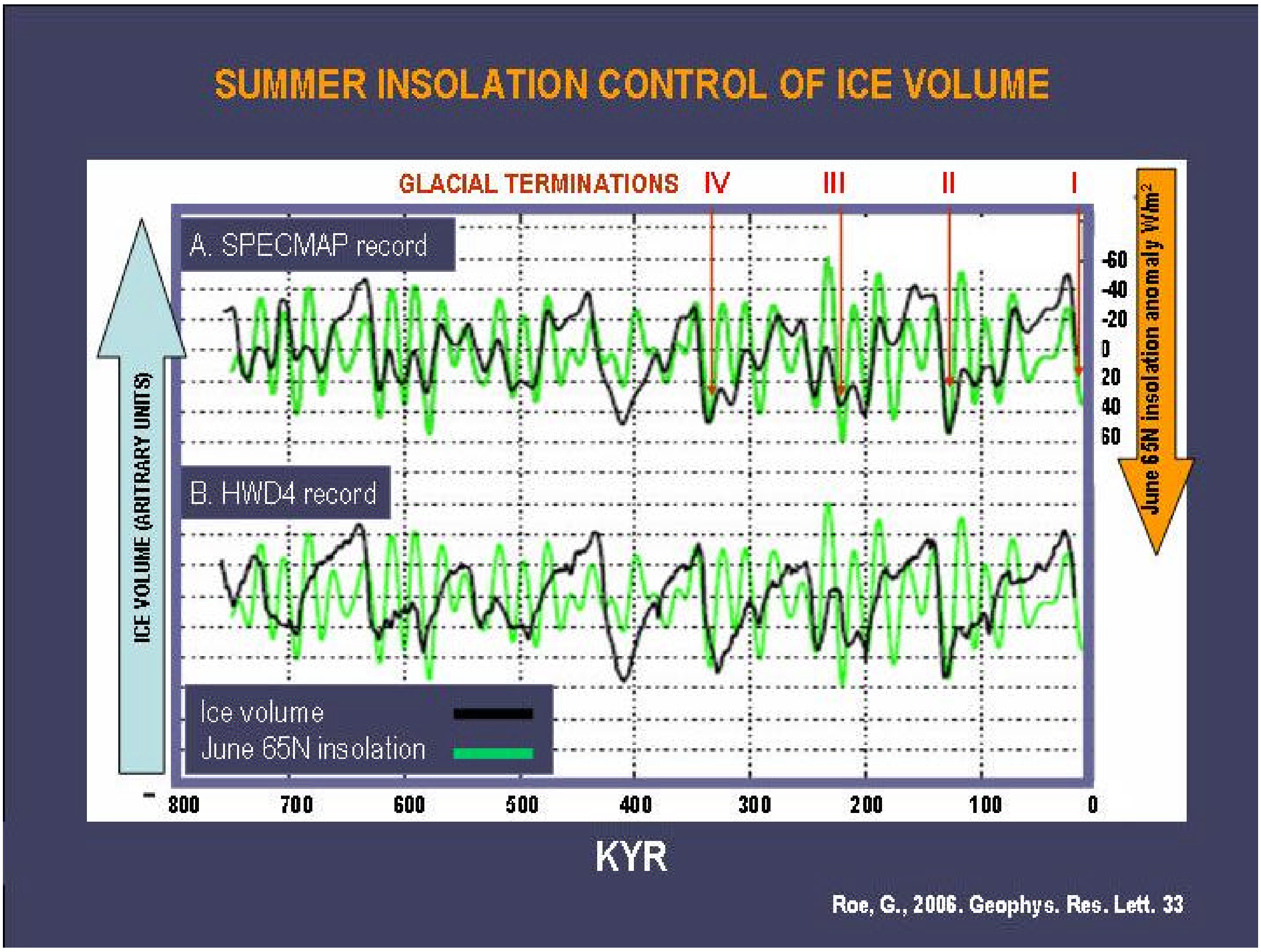

Figure 8. Variations in insolation and ice volumes during the last 740 kyr (after Roe, 2006)

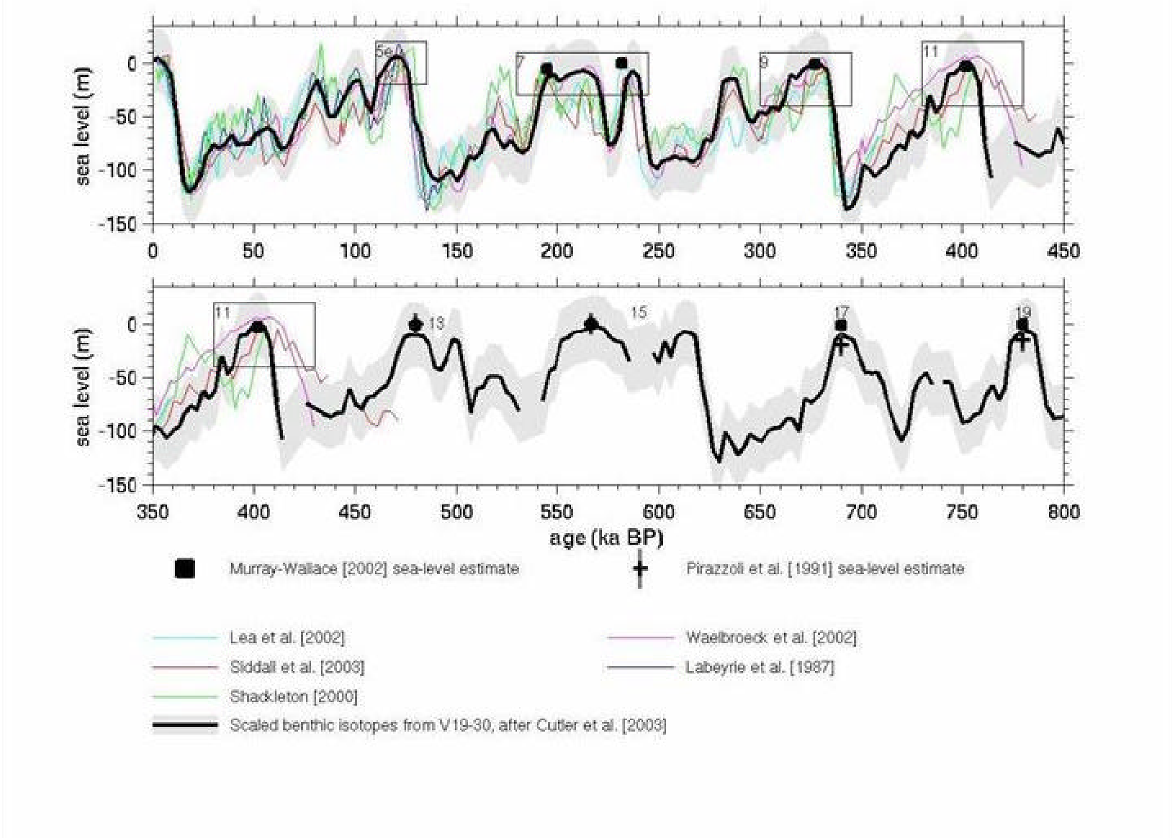

Figure 9. Estimates of sea level variations over the last 800 kyr (Siddall et al., 2005)

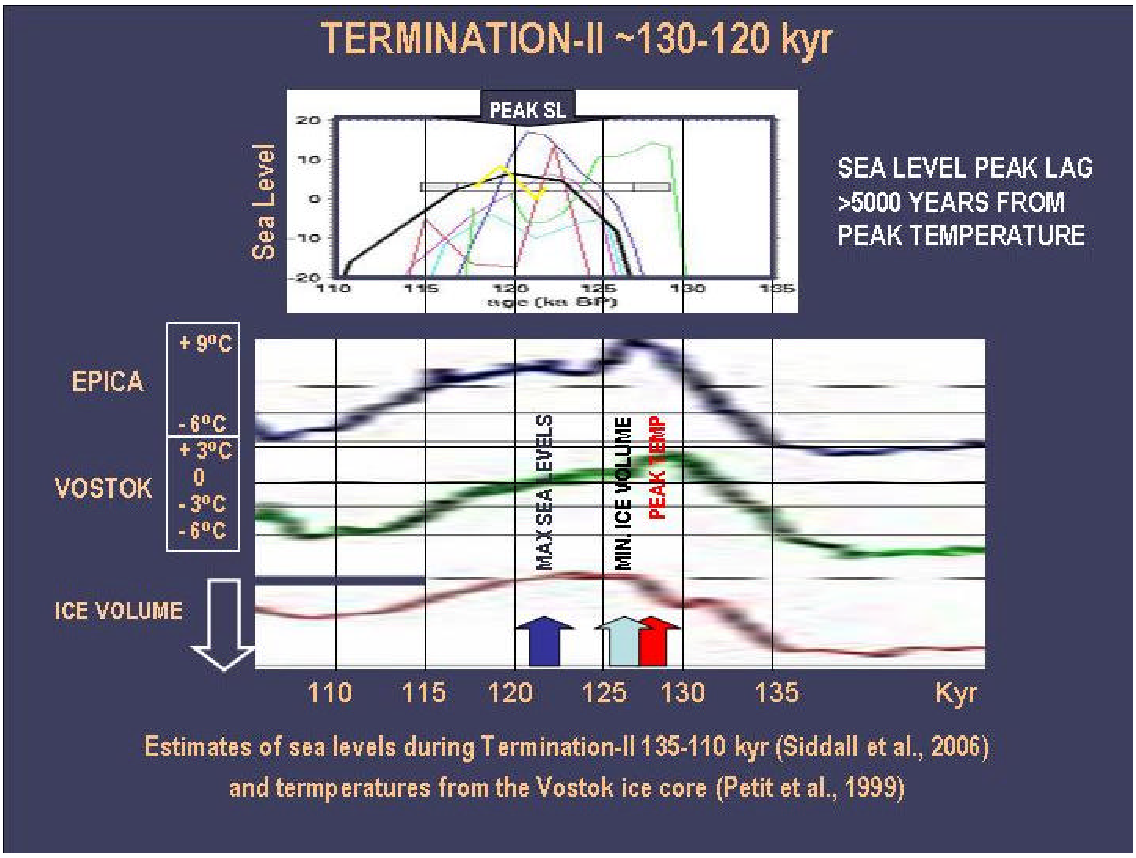

Figure 10. Comparisons between estimates of sea levels during glacial termination-II (135-122 kyr) (Sidall et al., 2006) and temperatures from the Vostok ice core (Petit et al., 1999).

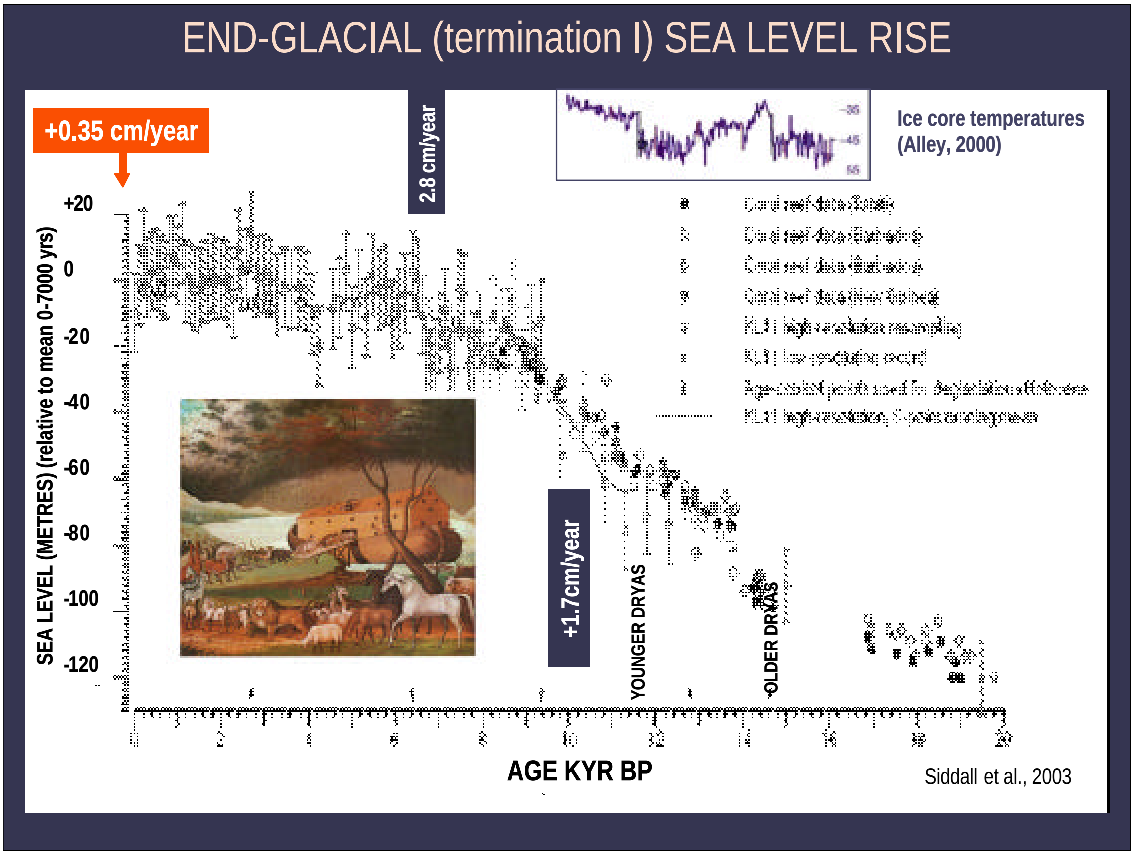

Figure 11. Sea-level reconstruction since the start of the Younger Dryas, based on the d18O record from core KL11 (188 44.50 N, 398 20.60 E) including error bars of +/-12 m. Chronology is based on calibrated AMS radiocarbon datings20. The record is zeroed to modern sea level by removing the mean KL11 record for the past 7 kyr. The point from the low-resolution record of KL11 reported in the LGM is based on intercalibrated benthic d18O data19. Black symbols are sea-level values obtained from coral reef studies21–24.

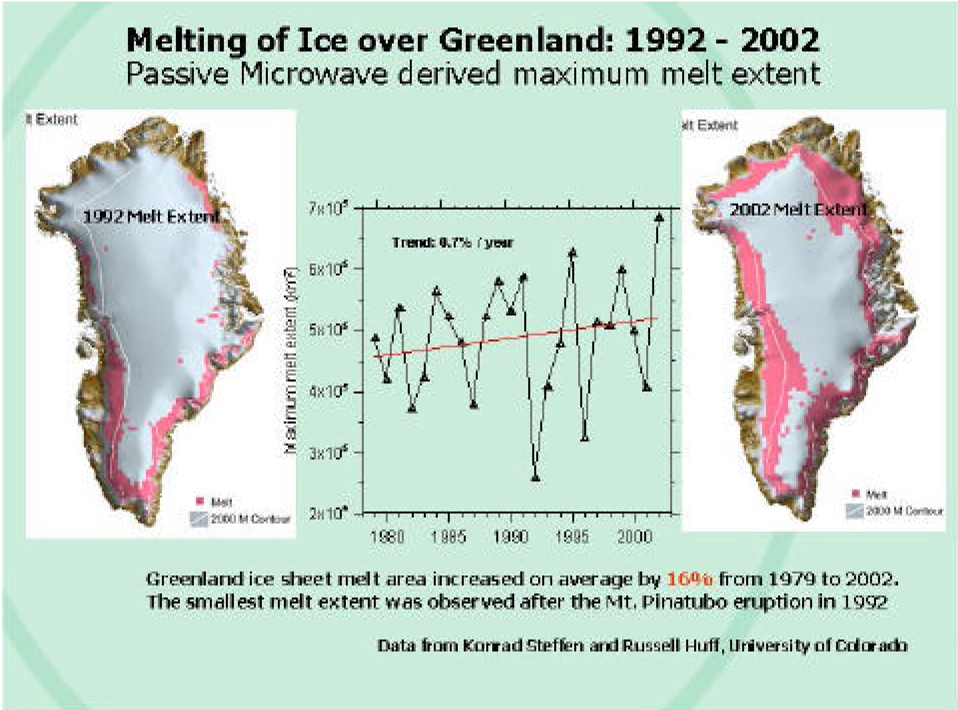

Figure 12. Greenland ice sheet melt area increase on average by 16% from 1979 to 2002. The smallest melt extent was observed after the Mt. Pinatubo eruption in 1992 (Steffen and Huff, 2002, 2004)

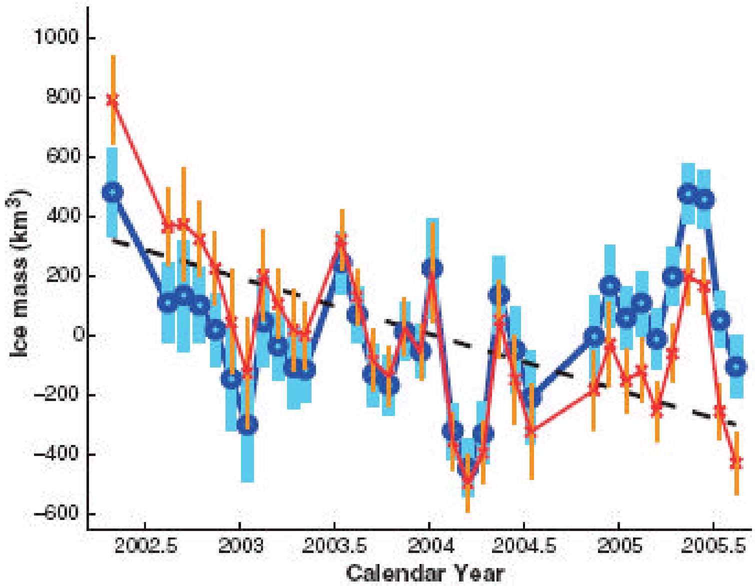

Figure 13. Antarctica: GRACE monthly mass solutions for the Antarctic ice sheet for April 2002 to August 2005 (Velicogna and Wahr, 2006) Fig. 2. GRACE monthly mass solutions for the Antarctic ice sheet for April 2002 to August 2005. Blue circles show results after removing the hydrology leakage. Red crosses show results after also removing the PGR signal. The latter represent our best estimates of the mass variability. The error bars include only the contributions from uncertainties in the GRACE gravity fields and represent 68.3% confidence intervals (13). Also shown is the linear trend that best fits the red crosses.

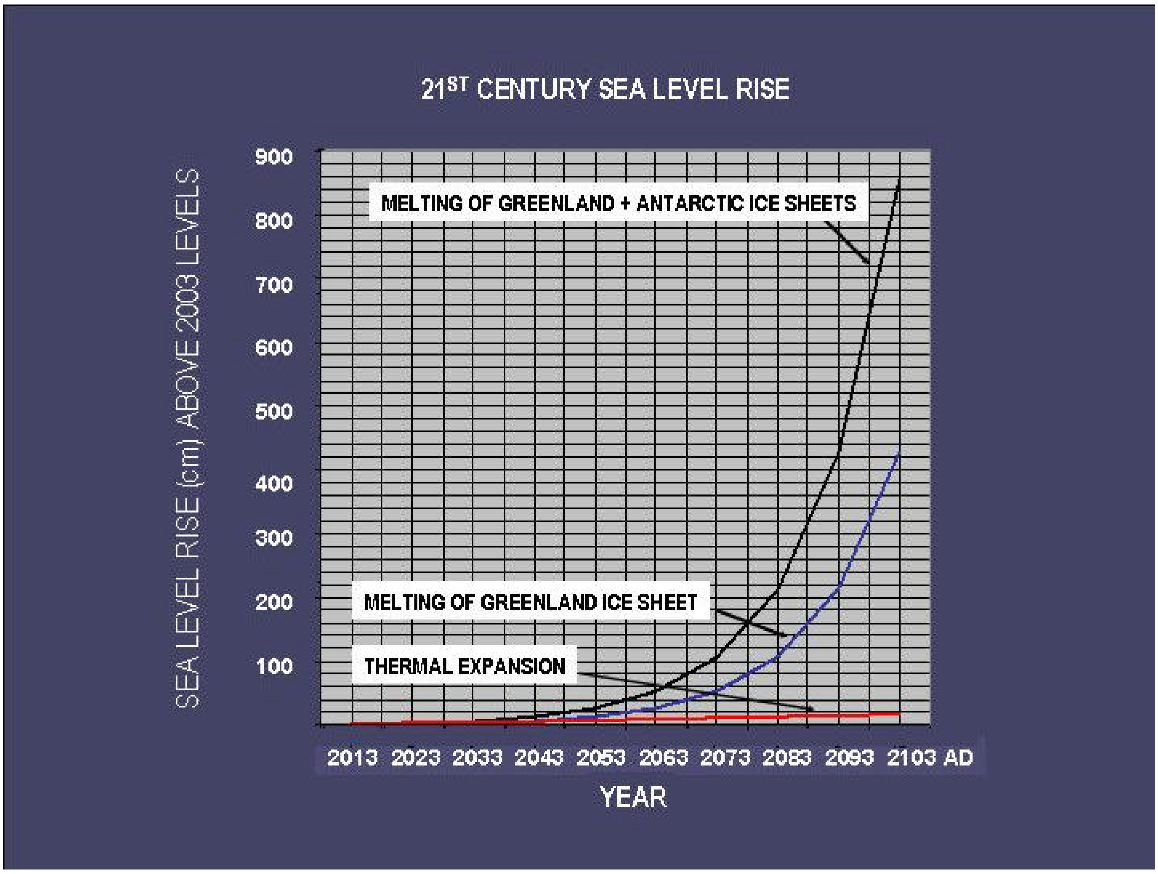

Figure 14. Projections of sea level rise through the 21st century based on Table 1A and B, assuming continual doubling per decade of Greenland and West Antarctic ice melting.

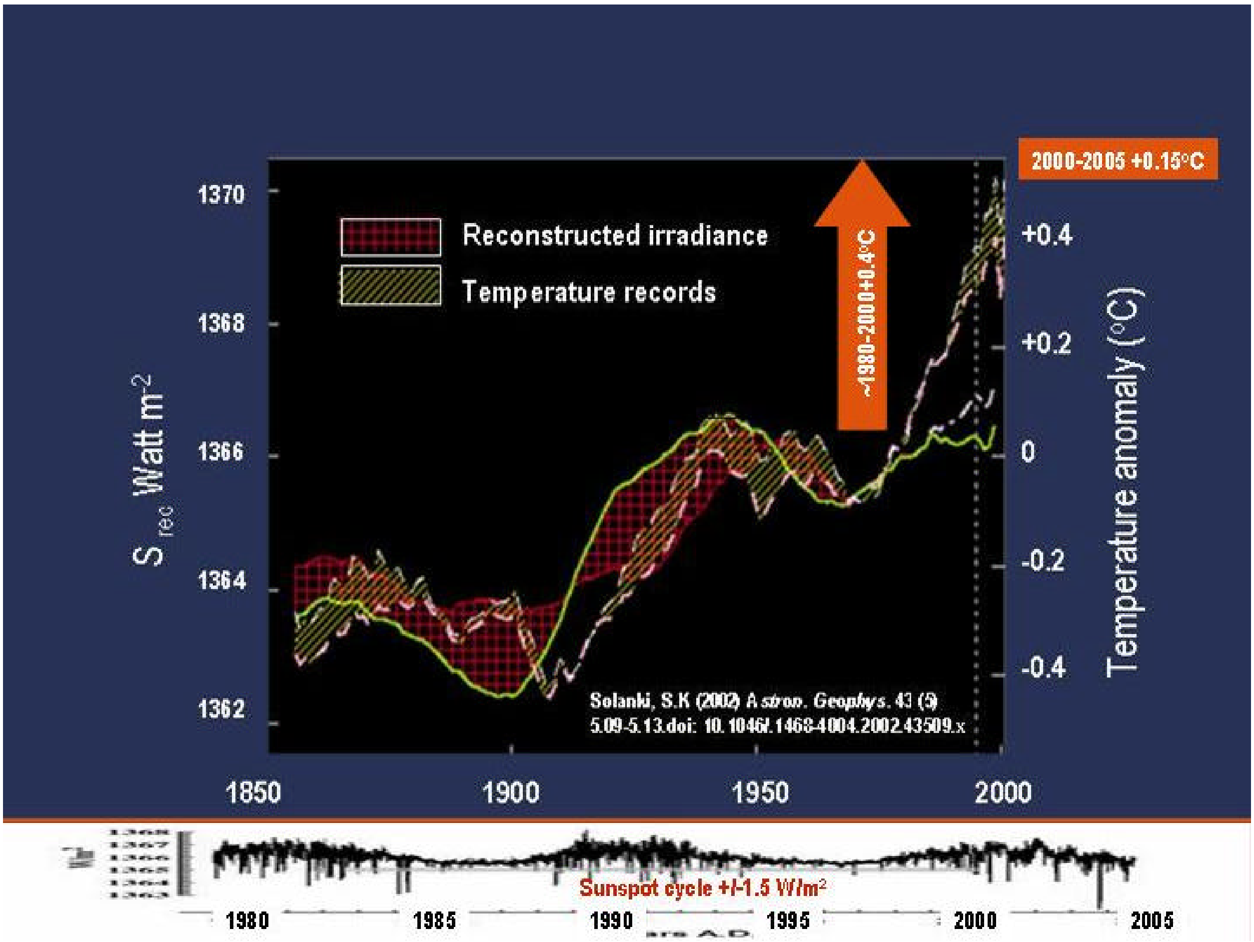

Figure 15. Variations in insolation, mean temperatures and radiative forcings through 1850-2000 (upper figure) and in sunspot activity during 1980-2005 (lower figure) (after Solanki, 2002)

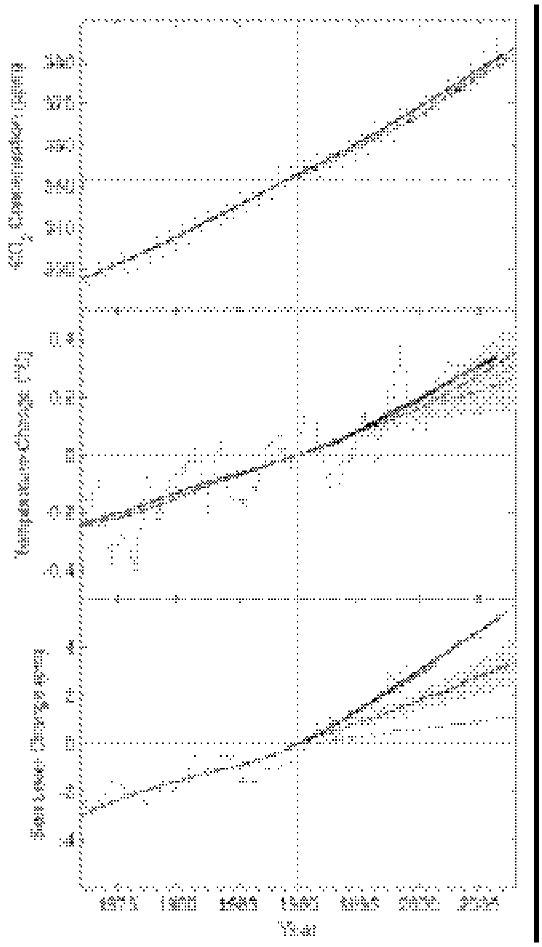

Figure 16. From Rahmstort et al., 2007: Changes in key global climate parameters since 1973, compared to the scenarios of the IPCC.

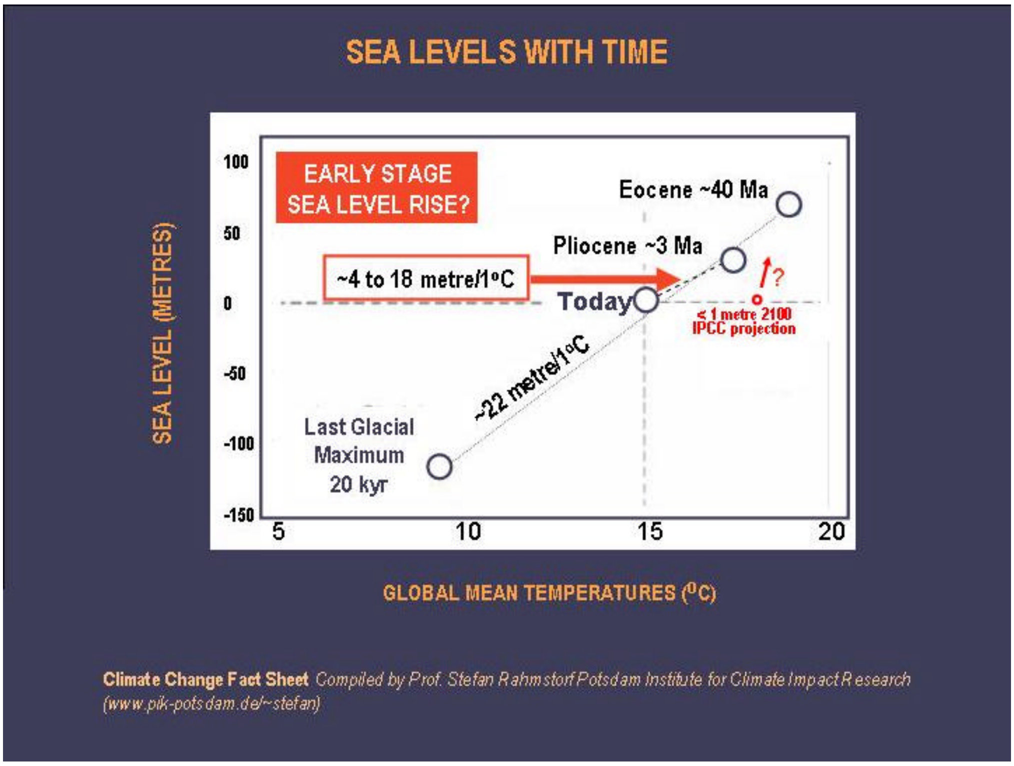

Figure 17. Mean global temperature and sea level (relative to today's) at different times in Earth's history, with the projection for the year 2100 (1m above today's sea level). For the long term a much higher sea-level rise probably must be expected than that predicted for 2100 (after Archer, 2006 and Rahmstort, 2007)